
Instructions. (15 points) Sections 15.3, 15.5, 15.6. Show relevant work for full credit.

(4^{pts}) 1.

Evaluate $\iint_D e^{-(x^2+y^2)} dA$, where D is the region between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

(6^{pts}) 2.

(a) (3 pts) Find the surface area of the part of the plane $z = 3 + x - 2y$ that lies above the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$.

(b) (3 pts) Set up, but do not evaluate, the surface area integral for $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 4$. Write your answer in polar coordinates.

(3^{pts}) **3.** Evaluate $\iiint_E z \, dV$, where E is the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

(2^{pts}) **4.** Set up, but do not evaluate, a triple integral for the volume of the solid in the first octant bounded by $z = 4 - y^2$ and $x = 3$.